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This supplement contains the following items:

1. BedMed Statistical Analysis Plan (The only version of the statistical analysis plan – created in advance of the interim analysis and submitted for publication on Dec 1, 2021)
2. Attestation and signatures of the Principal Investigator and Senior Statistician

*Note: The final version of the BedMed protocol, along with the final analysis plan, was published on Feb 24, 2022 as: *Garrison SR, Kolber MR, Allan GM, Bakal J, Green L, Singer A, Trueman DR, McAlister FA, Padwal RS, Hill MD, Manns B, McGrail K, O'Neill B, Greiver M, Froentjes LS, Manca DP, Mangin D, Wong ST, MacLean C, Kirkwood JE, McCracken R, McCormack JP, Norris C, Korownyk T. Bedtime versus morning use of antihypertensives for cardiovascular risk reduction (BedMed): protocol for a prospective, randomised, open-label, blinded end-point pragmatic trial. BMJ Open. 2022 Feb 24;12(2):e059711. doi: 10.1136/bmjopen-2021-059711. PMID: 35210352; PMCID: PMC8883279.*

BedMed Statistical Analysis Plan

Outcomes

Unless otherwise stated, all outcomes are recorded over the duration of the study.

Primary:

Major adverse cardiovascular events (MACE)

- Defined as first occurrence of either all-cause death or hospital admission / emergency department (ED) visit for acute coronary syndrome / myocardial infarction (MI), stroke, or congestive heart failure (CHF).

Secondary:

1. Each component of the primary outcome individually
2. All-cause hospitalization / ED visit
3. Long-term care (LTC) admission (i.e., to nursing home or assisted living facility)
4. Non-vertebral fracture
5. New glaucoma diagnosis
6. Cognitive decline at 18 months
 - Defined as ≥ 2 -point worsening in cognitive performance compared to baseline, as measured by the Short Blessed Test

Supplementary Safety Outcomes:

1. Vision
 - a. Vision self-reported as “much worse” compared to the last follow-up at any point, or “slightly worse” than the last follow-up, on 2 or more occasions (Note: vision is reported, every 6-months, as either “unchanged”, “slightly worse”, or “much worse” than the last follow-up)
2. Cognition
 - a. New “impairment consistent with dementia” at 18-months (Short Blessed Test newly ≥ 10) or new diagnosis of dementia at any point during follow-up
3. Symptomatic Hypotension
 - a. Self-reported light-headedness, or feeling faint without loss of consciousness, in the prior month
 - b. Self-reported fainting (loss of consciousness) in the prior month
 - c. Self-reported falling in the prior month
 - d. Hip fracture
(Note: at 6-weeks, 6-months, and every 6-months thereafter, participants are separately asked whether they have felt lightheaded, fainted, or fallen in the last month)
4. Nocturia
 - a. Self-reported change from baseline in the number of overnight urinations per week (at 6-weeks and 6-months)

- b. Self-reported nocturia burden in the prior month, recorded as no nocturia, or nocturia that is “no problem”, “minor problem”, or “major problem” (at 6-weeks and 6-months)

Cost:

1. Acute care costs (estimated from each hospital admission’s resource intensity weight and length of stay)*
2. Total cost of care (acute care costs + medication costs + physician billings)*

*All cost measures are derived entirely from administrative claims data, and not from self-report. If claims data is not available for some participants, they will be excluded from this analysis.

Exploratory:

1. Self-reported overall health score (via EQ-5D-5L) at 12-months

Process:

1. Proportion of BP medication doses taken at the allocated time at 6-months (twice daily medications being considered as ½ dose in the AM and ½ dose in the PM for this calculation)[†]
2. Sleep-time systolic blood pressure after 6-months (consecutive sample of 302 Alberta residents)[†]

[†]Although blinded to individual participant process outcomes, investigators are unblinded to the aggregated results for adherence to allocation time, and to the aggregated results from the 24-hour BP assessments. This allows for consideration of protocol alterations should the intervention appear poorly applied. Investigators are otherwise fully blinded to all trial outcomes.

Intension-to-treat assumptions

Loss to follow-up: If participants are lost to follow-up, but medical services continue to be recorded within administrative claims data, we will treat them as though they were still active in the study and censor survival data on the last date of medical services, or indication of death, whichever occurs later. If no such medical claims exist, data will be censored on the last day of successful telephone or e-mail follow-up.

Withdrawal: Participants withdrawing from the study are asked to allow us to continue to follow their administrative claims data. If they agree (as the majority do), we will continue to use administrative claims outcomes for those individuals as per the loss-to-follow-up description. If they do not agree, survival data will be censored on the date of withdrawal.

Missing data: For each analysis, we will either impute a value from subsequent or preceding follow-up visits, or exclude a participant from analysis. How we deal with missing data will be specific to each analysis and prioritize either minimizing bias, or

being conservative when bias is unavoidable (i.e. biasing against benefit, and towards harm, for the intervention).

Nonadherence: Nonadherence to allocation will not exclude participants from analysis unless the outcome of interest is a harm that only makes sense to assess while on-treatment (e.g. assessing how nocturia differs in diuretic users switched to bedtime, compared to nondiuretic users making the same switch).

Selecting regression covariates

Analyses of dichotomous outcomes will use a maximum of 1 covariate per 10 outcomes, and analyses of continuous outcomes will use a maximum of 1 covariate per 20 randomized subjects. The covariate list for each analysis is predefined in table 1, and all are measured at baseline. We will always use the maximum number of covariates possible, selected in the order given (i.e. we will not undertake stepwise addition or subtraction).

Table 1 Analysis Plan		
Outcome	Method	Covariates
Primary		
Major adverse cardiovascular events	Cox Proportional Hazards	Age, sex, frailty score*, current smoker, number of non-BP medications, Overall Health Score†, prior 6-months hospitalization, CHF, diabetes, CAD, stroke or TIA, CKD‡, dialysis, BMI > 35, BMI < 20, sleep apnea, exercise days§, province (4 variables)
Secondary		
All-cause mortality	Cox Proportional Hazards	Age, frailty score*, number of non-BP medications, Overall Health Score†, prior 6-months hospitalization, CHF, diabetes, CAD, CKD‡
Hospitalisation for stroke Hospitalisation for MI/ACS	Cox Proportional Hazards	Age, stroke or TIA, CAD, current smoker, sex, diabetes, exercise days§, BMI > 35
Hospitalisation for CHF	Cox Proportional Hazards	Age, CHF, CAD, diabetes, CKD‡
All-cause hospitalisation / ED visit	Cox Proportional Hazards	Age, sex, frailty score*, current smoker, number of non-BP medications, Overall Health Score†, prior 6-months hospitalization, CHF, diabetes, CAD, stroke, TIA, Short Blessed Test score, CKD‡, dialysis, BMI > 35, BMI < 20, COPD, province
Non-vertebral fracture	Cox Proportional Hazards	Age, Overall Health Score†, BMI, number of non-BP

		medications, frailty score*, stroke (not TIA), sex, CHF, exercise days§, TIA, prior 6-months hospitalization
LTC admission	Cox Proportional Hazards	Age ≥ 80, Short Blessed Test score, frailty score*
New glaucoma diagnosis	Cox Proportional Hazards	Age, diabetes, CAD or stroke or TIA, CHF, COPD, CKD‡, sleep apnea, BMI, exercise days§, Short Blessed Test Score
18-month cognitive decline	Poisson Regression	Age, sex, frailty score, number of non-BP medications, Overall Health Score†, CHF, stroke, TIA, COPD, BMI, exercise days§, province
Supplementary Safety		
Worsening of vision	Poisson Regression	Age, diabetes, CAD or stroke or TIA, CHF, COPD, CKD‡, Overall Health Score†
New impairment consistent with dementia	Poisson Regression	Age, sex, frailty score, number of non-BP medications, Overall Health Score†, CHF, stroke, TIA, COPD, BMI, exercise days§, province
Light-headedness in last Month Syncope in last month Falling in last month	Poisson Regression	Age, frailty score, number of non-BP medications, Overall Health Score†, CHF, stroke, TIA, sex, exercise days§, BMI, province
Hip fracture	Cox Proportional Hazards	Age, Overall Health Score†, BMI, number of non-BP meds, frailty score, stroke (not TIA), sex, CHF, exercise days§
Change in overnight urinations / week	Mann-Whitney or t-test	N/A
Nocturia a major burden	Fisher's Exact Test	N/A
Cost		
Acute care costs Total cost of care	Multiple Linear Regression	Age, sex, frailty score, current smoker, number of non-BP medications, Overall Health Score†, prior 6-months hospitalization, CHF, diabetes, CAD, stroke, TIA, Short Blessed Test score, CKD‡, dialysis, BMI > 35, BMI < 20, COPD, province
Exploratory		
Overall Health Score	Multiple Linear Regression	Age, sex, frailty score, current smoker, number of non-BP

		medications, prior 6-months hospitalization, CHF, diabetes, CAD, stroke, TIA, Short Blessed Test score, CKD‡, dialysis, BMI > 35, BMI < 20, COPD, province
*Score on physical frailty subscale of the Tilburg questionnaire; continuous 0-8 †From EQ-5D-5L; continuous 0-100 ‡Not including dialysis §“How many days in the past week have you exercised for 30 minutes or more, vigorously enough to raise your breathing rate?”; continuous 0-7		

Subgroup Analyses

We will repeat the primary outcome analysis for those with and without the following baseline characteristics: Age ≥ 75 , sex, physically frail (score ≥ 3 on physical frailty subscale of the Tilburg questionnaire), polypharmacy (≥ 5 medications), Overall Health Score ≤ 75 , resistant hypertension (≥ 3 BP-lowering medications), CHF, diabetes, CAD, stroke or TIA, sleep apnea, chronic kidney disease (with or without dialysis), sedentary (exercise 0 days per week).

Sensitivity Analyses

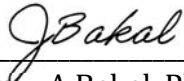
We will present, according to treatment group, the baseline characteristics of those whose data was censored due to withdrawal or loss to follow-up, and compare these characteristics to those who weren't censored in this way using Fisher's Exact Test.

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We the undersigned confirm that all analyses carried out for the BedMed randomized controlled trial make use of all available subjects and adhere completely to the detailed statistical analysis plan published as part of the BedMed protocol on Feb 24, 2022 and cited as:

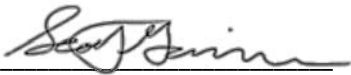
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